First Order Logic

- First-order logic is another way of knowledge representation in artificial intelligence. It is an extension to propositional logic.
- **First-order logic** is also known as **Predicate logic** or First-order predicate logic.
- Propositional logic assumes the world contain **facts**.
- First order logic assumes the world contains **Object, relations** and **function relation**.
- Objects: Person, numbers…
- Relations: brother of, part of,…
- Function Relation: father of, best friend,…

Predicate Logic Example

A sentence in **first-order logic** is written in the form Px or P(x), where P **is the predicate** and x **is the** subject(object), represented as a variable.

John is tall. **John is Subject and tall is predicate**

tall(John)

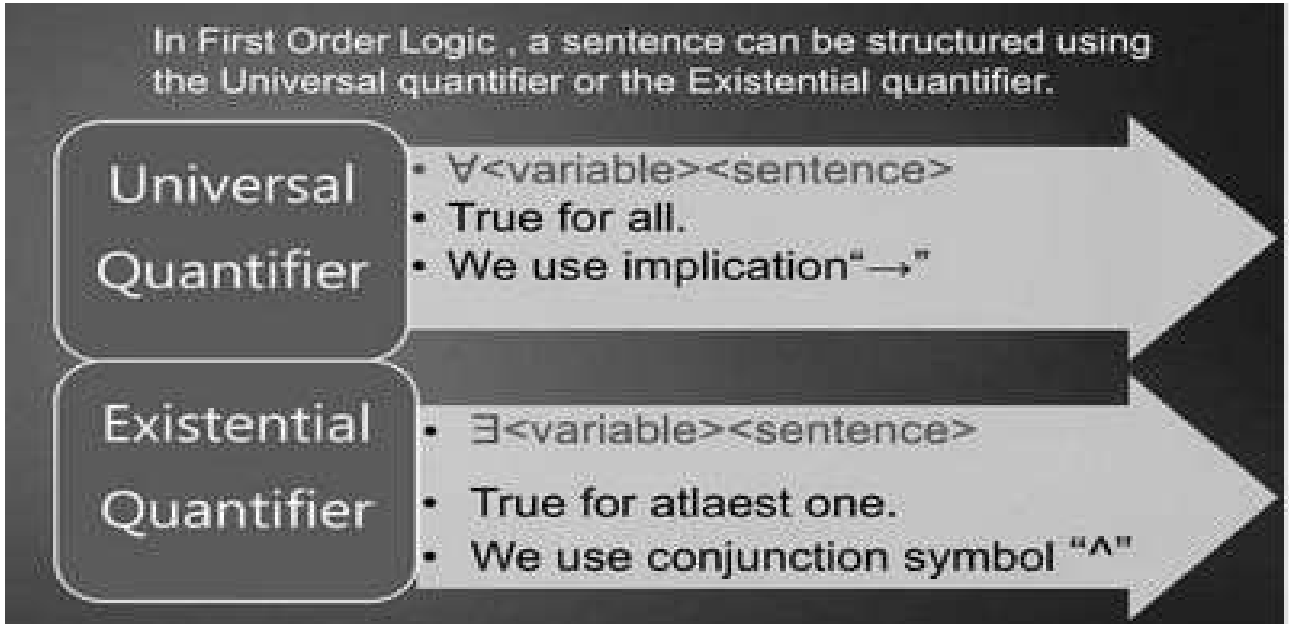
Arun and Vijay are friends.

friends(Arun , Vijay).

Basic Elements of Predicate Logic

Elements	Examples
Constant	1, 2, Pune, Ajay...
Variables	x, y, a , b,...
Predicates	Brother, Father,...
Function	Sqrt, LeftLegOf,...
Connectives	^, V, ¬, $\longrightarrow$ $\longleftrightarrow$
Equality	==
Quantifier	∀, ∃

Quantifiers



Quantifiers

- **Quantifiers**

  Universal quantification ∀ (or ***forall***)

  – (∀x)P(x) means that P holds for **all** values of x in the domain associated with that variable.

  – E.g., (∀x) dolphin(x) => mammal(x)

  (∀x) human(x) => mortal(x)

  – Universal quantifiers often used with "implication (=>)" to form "rules" about properties of a class

  (∀x) student(x) => smart(x) (All students are smart)

  – Often associated with English words "all", "everyone", "always", etc.

  – You rarely use universal quantification to make blanket statements about every individual in the world (because such statement is hardly true)

  (∀x)student(x)^smart(x) means everyone in the world is a student and is smart.

Quantifiers

Existential quantification ∃

- (∃x)P(x) means that P holds for **some** value(s) of x in the domain associated with that variable.
- E.g., (∃x) mammal(x) ^ lays-eggs(x)

  (∃x) taller(x, Fred)

  (∃x) UMBC-Student (x) ^ taller(x, Fred)
- Existential quantifiers usually used with "^ (and)" to specify a list of properties about an individual.

  (∃x) student(x) ^ smart(x) (there is a student who is smart.)
- A common mistake is to represent this English sentence as the FOL sentence:

  (∃x) student(x) => smart(x)

  It also holds if there no student exists in the domain because

  student(x) => smart(x) holds for any individual who is not a student.
- Often associated with English words "someone", "sometimes", etc.

Scopes of Quantifiers

Scopes of quantifiers

- Each quantified variable has its scope
  - $(\forall x)[human(x) \Rightarrow (\exists y)\ [human(y) \wedge father(y, x)]$
  - All occurrences of x within the scope of the quantified x refer to the same thing.
  - Use different variables for different things
- Switching the order of universal quantifiers does not change the meaning:
  - $(\forall x)(\forall y)P(x,y) \Leftrightarrow (\forall \mathbf{y})(\forall x)P(x,y)$, can write as $(\forall x,y)P(x,y)$
- Similarly, you can switch the order of existential quantifiers.
  - $(\exists \mathbf{x})(\exists y)P(x,y) \Leftrightarrow (\exists y)(\exists \mathbf{x})P(x,y)$
- Switching the order of universals and existential does change meaning:
  - Everyone likes someone: $(\forall x)(\exists y)likes(x,y)$
  - Someone is liked by everyone: $(\exists y)(\forall x)\ likes(x,y)$

Sentences are built from terms and atoms

Sentences are built from terms and atoms

- A **term** (denoting a individual in the world) is a constant symbol, a variable symbol, or a function of terms.
- An **atom** (atomic sentence) is a predicate P(x[1], ..., x[n])
  - Ground atom: all terms in its arguments are ground terms (does not involve variables)
  - A ground atom has value true or false (like a proposition in PL)
- A **literal** is either an atom or a negation of an atom
- A **sentence** is an atom, or,
  - ~P, P v Q, P ^ Q, P => Q, P <=> Q, (P) where P and Q are sentences
  - If P is a sentence and x is a variable, then (∀x)P and (∃x)P are sentences
- A **well-formed formula (wff)** is a sentence containing no "free" variables. i.e., all variables are "bound" by universal or existential quantifiers.

  (∀x)P(x,y) has x bound as a universally quantified variable, but y is free.

Variable

Free and Bound Variables

- The quantifiers interact with variables which appear in a suitable way. There are two types of variables in First-order logic which are given below:
- **Free Variable:** A variable is said to be a free variable in a formula if it occurs outside the scope of the quantifier.

Example: $\forall x\ \exists(y)[P(x, y, z)]$, where z is a free variable.

- **Bound Variable:** A variable is said to be a bound variable in a formula if it occurs within the scope of the quantifier.

Example: $\forall x\ [A(x)\ B(y)]$, here x and y are the bound variables.

BNF for FOL

A BNF for FOL Sentences

```
S := <Sentence> ;
<Sentence> := <AtomicSentence> |
          <Sentence> <Connective> <Sentence> |
          <Quantifier> <Variable>,... <Sentence> |
          ~ <Sentence> |
          "(" <Sentence> ")";
<AtomicSentence> := <Predicate> "(" <Term>, ... ")" |
                    <Term> "=" <Term>;
<Term> := <Function> "(" <Term>, ... ")" |
          <Constant> |
          <Variable>;
<Connective> := ^ | v | => | <=>;
<Quantifier> := ∃ | ∀;
<Constant> := "A" | "X1" | "John" | ... ;
<Variable> := "a" | "x" | "s" | ... ;
<Predicate> := "Before" | "HasColor" | "Raining" | ... ;
<Function> := "Mother" | "LeftLegOf" | ... ;
<Literal> := <AutomicSetence> | ~ <AutomicSetence>
```

Some Examples of FOL

1. All kings are person.

$\forall x$: King(x) $\Longrightarrow$ person(x)

2. No-one likes noodles.

$\forall x$: person(x) $\Longrightarrow$ $\neg$ likes(x, noodles)

3. Some dog is a pet.

$\exists x$: (dog(x) $\wedge$ pet(x))

4. Every student pass or fail.

$\forall x$: (student(x) $\Longrightarrow$ (pass(x) $\vee$ fail (x)))

5. If x is a parent of y then x is older than y.

$\forall x \forall y$ Parent(x,y) $\Longrightarrow$ older(x,y)

Properties of Quantifiers

- $\forall x\ \forall y$ is the same as $\forall y\ \forall x$
- $\exists x\ \exists y$ is the same as $\exists y\ \exists x$
- $\exists x\ \forall y$ is not the same as $\forall y\ \exists x$
- $\exists x\ \forall y$ loves(x,y)- "There is a person who loves everyone in the world"
- $\forall y\ \exists x$ loves(x,y)-"Everyone in the world is loved by at least one person"
- Quantifier duality: Each can be expressed using the other.
- $\forall x$ likes(x, IceCream) $\neg\ \exists x\ \neg$likes(x,Icecream)
- $\exists x$ plays(x, Cricket) $\neg\ \forall x\ \neg$plays(x, Cricket)

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Translating English to FOL

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- **Every gardener likes the sun.**
  $(\forall x)$ gardener(x) => likes(x,Sun)
- **Not Every gardener likes the sun.**
  ~($(\forall x)$ gardener(x) => likes(x,Sun))
- **You can fool some of the people all of the time.**
  $(\exists x)(\forall t)$ person(x) ^ time(t) => can-be-fooled(x,t)
- **You can fool all of the people some of the time.**
  $(\forall x)(\exists t)$ person(x) ^ time(t) => can-be-fooled(x,t)
  (the time people are fooled may be different)
- **You can fool all of the people at some time.**
  $(\exists t)(\forall x)$ person(x) ^ time(t) => can-be-fooled(x,t)
  (all people are fooled at the same time)
- **You can not fool all of the people all of the time.**
  ~($(\forall x)(\forall t)$ person(x) ^ time(t) => can-be-fooled(x,t))
- **Everyone is younger than his father**
  $(\forall x)$ person(x) => younger(x, father(x))

Translating English to FOL

- **All purple mushrooms are poisonous.**
  (∀x) (mushroom(x) ^ purple(x)) => poisonous(x)
- **No purple mushroom is poisonous.**
  ~(∃x) purple(x) ^ mushroom(x) ^ poisonous(x)
  (∀x) (mushroom(x) ^ purple(x)) => ~poisonous(x)
- **There are exactly two purple mushrooms.**
  (∃x)(Ey) mushroom(x) ^ purple(x) ^ mushroom(y) ^ purple(y) ^ ~(x=y) ^
  (∀z) (mushroom(z) ^ purple(z)) => ((x=z) v (y=z))
- **Clinton is not tall.**
  ~tall(Clinton)
- **X is above Y if X is directly on top of Y or there is a pile of one or more other objects directly on top of one another starting with X and ending with Y.**
  (∀x)(∀y) above(x,y) <=> (on(x,y) v (∃z) (on(x,z) ^ above(z,y)))

Syntax of FOL

Syntax of FOL

- **Predicates:** P(x[1], ..., x[n])
  - P: **predicate name**; (x[1], ..., x[n]): **argument list**
  - A special function with range = {T, F};
  - Examples: human(x), /* x is a human */
    father(x, y) /* x is the father of y */
  - When all arguments of a predicate is assigned values (said to be **instantiated**), the predicate becomes either true or false, i.e., it becomes a proposition. Ex. Father(Fred, Joe)
  - A predicate, like a membership function, defines a set (or a class) of objects
- **Terms** (arguments of predicates must be terms)
  - **Constants** are terms (e.g., Fred, a, Z, "red", etc.)
  - **Variables** are terms (e.g., x, y, z, etc.), a variable is **instantiated** when it is assigned a constant as its value
  - **Functions** of terms are terms (e.g., f(x, y, z), f(x, g(a)), etc.)
  - A term is called a **ground** term if it does not involve variables

FOL

Practice

1. All birds fly.
2. Every man respects his parent.
3. Some boys play cricket.
4. Not all students like both Mathematics and Science.
5. Only one student failed in Mathematics.

FOL

1. $\forall x$ bird(x) $\rightarrow$fly(x).
2. $\forall x$ man(x) $\rightarrow$ respects (x, parent).
3. $\exists x$ boys(x) $\rightarrow$ play(x, cricket).
4. $\neg\forall$ (x) [student(x) $\rightarrow$ like(x, Mathematics) $\wedge$ like(x, Science)].
5. $\exists$(x) [student(x) $\rightarrow$ failed (x, Mathematics) $\wedge\forall$ (y) [$\neg$(x==y) $\wedge$ student(y) $\rightarrow$ $\neg$failed (y, Mathematics)].

FOL

Exercise

1. John likes Cricket OR football
2. John lives in a house and the color of the house is green.
3. If car belongs to john then it is green.
4. John did not write Ramayana.
5. All elephants are gray.
6. Some courses in CSE are easy

Solutions of Exercise of FOL

- Likes(John, Cricket) V likes(John, football)
- Lives(john, house) ∧ colour (house, green)
- Belongs(car, john) → colour (car, green)

- ¬Write(john, ramayana)

- All elephants are gray
  X:elephant(x) →colour(x, gray)
- Some courses in CSE are easy
  ∃X: Course(x) ∧ easy(x, CSE)

Exercise of FOL

- Venkatesh only likes easy courses.
- Science courses are not easy.
- All the courses in the arts department are easy.
- B.Sc is a science course

Exercise of FOL

- Venkatesh only likes easy courses

  X:easycourse(x) →likes(venkatesh,X)
- Science courses are not easy

  x:science_course(x) →¬easy_course(x)
- All the courses in the arts department are easy

  x:course(x)∧arts(x) → easy(x)

- B.Sc is a science course

  science_course(B.Sc)

Higher Order Logic

Higher order logic

- FOL only allows to quantify over variables, and variables can only range over objects.
- HOL allows us to quantify over relations
- Example: (quantify over functions)

  "two functions are equal iff they produce the same value for all arguments"

  $\forall f \forall g\ (f = g) \Leftrightarrow (\forall x\ f(x) = g(x))$
- Example: (quantify over predicates)

  $\forall r\ \text{transitive}(\ r\ ) \Rightarrow (\forall xyz)\ r(x,y) \wedge r(y,z) \Rightarrow r(x,z))$
- More expressive, but undecidable.